

ABSTRACT

Predictive Maintenance Of Aircraft Engines

In the past if a mechanic was servicing a machine and found unusual visual or acoustic behaviour in a certain part, the machine may be shut down before breaking and the part was exchanged. That is already predictive maintenance. The new thing today is that machines and algorithms can perform this task 24/7 based on smart algorithms, more data and automated alerts for detecting irregularities. In predictive maintenance scenarios, data is collected over time to monitor the state of equipment. As in most ML projects, there is a need for enough historical data to understand previous failures.

Applications of predictive maintenance are massive cost savings mainly due to decreased downtime, avoidance of related machinery failure due to connected parts, reduction of storage cost for spare parts. Predictive maintenance determines the condition of equipment and predicts when maintenance should be performed. The goal is to find patterns that can help to predict and ultimately prevent failures.

Failure prediction is a major topic in predictive maintenance in many industries. Aircrafts are particularly interested in predicting equipment failures in advance so that they can enhance their operations and reduce delays. The aim of predictive maintenance in aviation is first to predict when a component failure might occur, and secondly, to prevent the occurrence of the failure by performing maintenance. Observing engine's health and condition through sensors and telemetry data is assumed to facilitate this type of maintenance by using machine learning algorithms.

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